

Harvey Zhu

713-380-8227 | hv639@cornell.edu | [linkedin.com/in/zhuharvey](https://www.linkedin.com/in/zhuharvey) | github.com/harveyzhu27 | harvey-zhu.vercel.app

EDUCATION

Cornell University

Ithaca, NY

Bachelor of Arts in Computer Science, Minor in Artificial Intelligence

Expected May 2027 / GPA: 3.7

Relevant Coursework:

- Data Structures, Functional Programming, Analysis of Algorithms, Operating Systems, Computer Graphics, Game Design, Machine Learning, Computer System Organization, Backend Development, Android Development, Discrete Structures, Linear Algebra, Probability Models and Inference

Teaching Assistant (Data Structures and Functional Programming)

Aug 2025 – Present

- Leading discussion sessions of 40+ students and hold weekly office hours to support 30+ students.
- Collaborating with professor to design assignments, projects and exams, assisting in Ed Discussion forums.

Research Assistant (Cornell SC Johnson College of Business)

Sept 2025 – Present

- Building a full-stack Beer Game with backend systems to model supply chain and frontend for interactive gameplay
- Developing AI agents and integrating LLMs to analyze strategies and emulate human-like decision making.

EXPERIENCE

Hawl Technologies | *Software Engineering Intern, Technical Lead*

Aug 2025 – Sept 2025

- Directed development of an AI-powered enterprise search platform that reduced hallucinations by combining retrieval from 140+ trusted sources with multi-stage validation, delivering higher accuracy and reliability for users.
- Implemented secure RESTful API with dual authentication (JWT + user-provided keys), real-time streaming, and usage analytics, improving client adoption and reducing credential management risks.
- Engineered MongoDB Atlas schema with strict JSON validation, compound indexing, and encryption-at-rest.

Teidore | *Software Engineering Intern*

Jun 2025 – Sept 2025

- Deployed automated CI/CD on AWS Amplify, EC2 and CloudWatch for monitoring and prompt release cycles.
- Designed a Python-based ETL pipeline migrating retailer product data from AWS S3 into PostgreSQL RDS, improving processing throughput to 5,000+ records/minute while ensuring data validation across 1M+ records.
- Identified performance bottlenecks in an unoptimized e-commerce schema and redesigned it with hierarchical categories and materialized views, cutting query latency by 40% and supporting scalable product queries.
- Built AI/ML pipelines with BERT and Ollama for auto-classification, improving categorization accuracy by 125%.

Gilded Flame | *Project Lead, Programmer, Artist*

Jan 2025 – May 2025

- Faced with a one-semester deadline to build a game featuring dynamic lighting, AI-driven enemies, and complex puzzle logic, led a 9-person team of engineers and artists to deliver a fully functional game by showcase day.
- Guided the team through production challenges by reallocating resources, resolving frontend and backend conflicts, keeping milestones on track and a successful presentation to 150+ attendees at Cornell's GDIAC Showcase.

PROJECTS

Database Management CLI Tool | *TypeScript, PostgreSQL, Next.js, Drizzle ORM*

[GitHub](#)

- Built a CLI tool to generate database schemas, migrations, and routes, auto-updating files and frontend integration
- Automated backend with Drizzle ORM + PostgreSQL for database management and API route generation
- Integrated OpenAI NLP to parse natural language prompts into structured DB operations and project updates

Voice-Enabled Support Chatbot | *TypeScript, Next.js, React, Pinecone, Vapi, WebRTC*

[GitHub](#)

- Built AI-supported assistant with GPT-3.5, semantic search via Pinecone embeddings, and confidence scoring
- Implemented real-time voice interaction (Vapi, WebRTC) with modern Next.js/React frontend and Tailwind
- Developed transcript processing pipeline enabling context-aware FAQ retrieval and accurate AI-generated responses

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, HTML/CSS/JS, Python, SQL, OCaml, C, C++, Kotlin, Rust

Frameworks/Tools: React, Next.js, FastAPI, Node.js/Express, Firebase, Docker, Supabase, PostgreSQL, MongoDB, AWS (S3, RDS, EC2, Amplify), Vercel, Render, TensorFlow.js, BERT, Pinecone, Ollama, Electron, JetPack Compose

Developer Tools: Git, VS Code, IntelliJ, Eclipse, Postman, Ubuntu/Linux, CI/CD (GitHub Actions, AWS Amplify)